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PROJECT AND TEAM INFORMATION

Project Title

HyperVision: A Resource-Aware DBMS–OS Laboratory Using Xen and Virtual Machine Introspection

The Hypervisors *Team # OSDBMS-V-2026-T271*	
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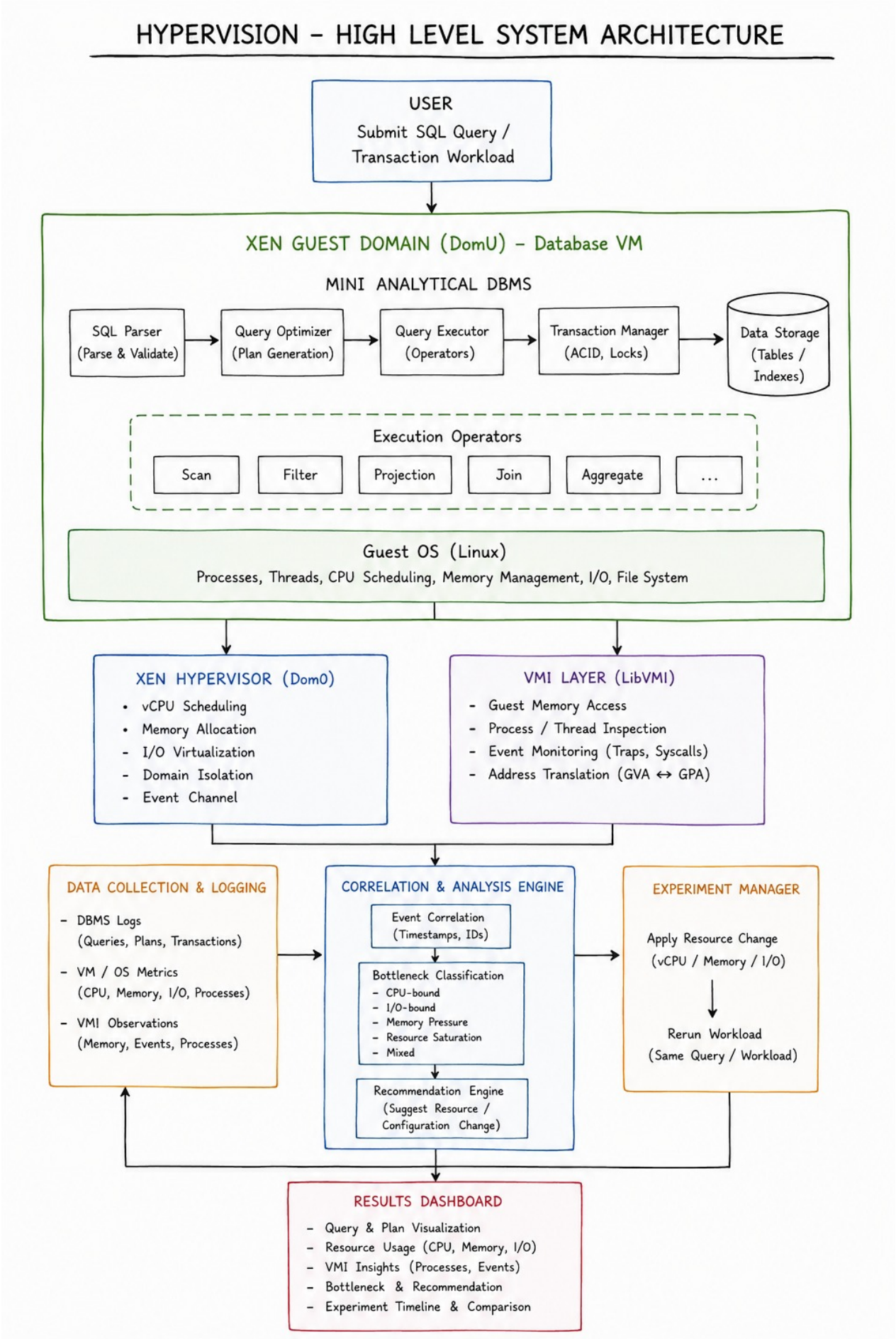
PHASE-II PROJECT PROGRESS DESCRIPTION

Project Abstract

When a database query runs slow,the problem is not always in the query.It can also happen because of CPU,memory or threads of the computer.These things are mostly checked separately, so finding the real problem can be difficult.In HyperVision,we made a small database system using Python.It takes a SQL query,makes a plan and runs it using scan,filter,join and group operations.While running the query,our program checks CPU and memory usage.Then it uses some simple rules to find if the query is CPU-bound, memory-limited,I/O-bound or mixed.After this,it gives a simple sugestion for improvment.We also run the same query again with different number of threads and compare both results.A Flask web dashboard shows the query plan,resource usage,result and experiment history.We also made a small lock manager which can detect deadlock.For real Xen setup,special hardware is needed,so we simulated the Xen and VMI part using code.In future,this simulated part can be replaced with real Xen and LibVMI.

Updated Project Approach and Architecture

In Phase 1,we planned to run the database inside a Xen DomU.But real Xen needs special hardware,so in Phase 2 we made the backend run on a normal laptop.The Xen and VMI part is simulated using code.In this phase,we completed only the backend and run it from the terminal.The dashboard and its connection with backend is planned for the next phase.The user gives a SQL query,and the experiment manager gives it an experiment ID.Then it run the query through our mini DBMS.It includes parser,optimizer,scan,filter,join and group operations.While the query is runing,the resource monitor checks CPU and memory usage.The VMI adapter also gives simulated guest events.The database steps and resource reading are joined together in one timeline.Then some simple rules are used to find if the run is CPU-bound,memory pressure,I/O-bound or mixed.After this,a recomender gives a suggestion for improvment.The same query is run again with different number of threads and both results are compared.All the results are saved in a log file.The VMI adapter has a fixed interface,so in future we can replace the simulated part with real Xen and LibVMI without changing the other parts of the project.


HYPERVISION – HIGH LEVEL SYSTEM ARCHITECTURE
USER
Submit SQL Query / Transaction Workload
XEN GUEST DOMAIN (DomU) – Database VM
MINI ANALYTICAL DBMS
SQL Parser (Parse & Validate)
Query Optimizer (Plan Generation)
Query Executor (Operators)
Transaction Manager (ACID, Locks)
Data Storage (Tables / Indexes)
Execution Operators
Scan
Filter
Projection
Join
Aggregate
...
Guest OS (Linux)
Processes, Threads, CPU Scheduling, Memory Management, I/O, File System
XEN HYPERVISOR (Dom0)
• vCPU Scheduling
• Memory Allocation
- I/O Virtualization
- Domain Isolation
- Event Channel
VMI LAYER (LibVMI)
- Guest Memory Access
- Process / Thread Inspection
- Event Monitoring (Traps, Syscalls)
- Address Translation (GVA ↔ GPA)
DATA COLLECTION & LOGGING
- DBMS Logs (Queries, Plans, Transactions)
- VM / OS Metrics (CPU, Memory, I/O, Processes)
- VMI Observations (Memory, Events, Processes)
CORRELATION & ANALYSIS ENGINE
Event Correlation (Timestamps, IDs)
Bottleneck Classification
- CPU-bound
- I/O-bound
- Memory Pressure
- Resource Saturation
- Mixed
Recommendation Engine (Suggest Resource / Configuration Change)
EXPERIMENT MANAGER
Apply Resource Change (vCPU / Memory / I/O)
Rerun Workload (Same Query / Workload)
RESULTS DASHBOARD
- Query & Plan Visualization
- Resource Usage (CPU, Memory, I/O)
- VMI Insights (Processes, Events)
- Bottleneck & Recommendation
- Experiment Timeline & Comparison

Tasks Completed

Task Completed	Team Member
1. Built the mini DBMS.	Naman Garg
2. Added Resource Monitor.	Mahesh Singh
3. Added a lock manager that detects deadlock.	Karishma Rastogi
4. Built Experiment Manager.	Naman Garg
5. Tested the backend from the terminal.	Mahesh Singh

Challenges/Roadblocks

1. Xen needs special hardware – Xen needs special hardware to run properly.A real Xen hypervisor needs a dedicated Linux machine with hardware virtualization,so we simulated the Xen and VMI part using code.
2. Changing vCPUs was not possible – Changing vCPUs was not possible on our laptop.So,we changed the number of worker threads instead to show how more resources can affect the query.
3. Parsing SQL was difficult – Handling WHERE, JOIN and GROUP BY together in one parser was difficult. So,we support only some limited types of SQL queries.
4. Fast queries gave few readings – Some queries was running very fast,so only few CPU and memory readings was collected.Because of this,the classifier sometimes gives Mixed.Using larger sample data helped to get better readings.
5. Threshold values needed testing – The CPU and memory threshold values needed some trial and error.We tested different values on our laptops to make the classifier work better.

Tasks Pending

Task Pending	Team Member (to complete the task)
Connect backend with dashboard using Flask.	Naman Garg
Set up real Xen and DomU on Linux.	Mahesh Singh
Install LibVMI and use real VMI part.	Mahesh Singh
Control vCPU and memory instead of threads.	Karishma Rastogi
Add more SQL queries and join types.	Naman Garg
Complete the final report and project demo.	Karishma Rastogi

Project Outcome/Deliverables

HyperVision is currently terminal-based and made using Python.It is a small DBMS which can read some SQL queries,make a query plans and run operations like scan,filter,projection,join,group and sort.We also added a lock manager for finding deadlock.The system check CPU and memory usage using a resource monitor.It also have a simulated VMI adapter for showing guest event.The experiment manager gives an experiment ID,make a timeline and find the problem as CPU-bound,memory pressure,I/O-bound or mixed.It also give some simple sugestions.The same query can be run again with different number of worker threads and both results can be compared.All results are saved in a log file.For now,the project is terminal-based.The web dashboard,real Xen/LibVMI and more SQL features will be added in the next phase.

Progress Overview

The project is currently at the terminal-based backend stage.The main backend parts are completed,like mini DBMS,SQL parser,query plan,scan,filter,projection,join group and sort operators.We also made lock manager,resource monitor,simulated VMI adapter and experiment manager.Bottleneck classifier,recomendations and thread-based rerun are also completed.The backend is tested by running experiments from the terminal.The project is little behind from the original plan because real Xen needs special hardware and suported Linux environment.So,Xen and VMI is curently simulated in code.We are also using worker threads instead of changing DomU vCPUs.The web dashboard is not made or connected yet.Real Xen with DomU,LibVMI integration,more SQL support,more experiment and classifier tuning are still pending.The final report and demo is also remaining.

Codebase Information

Github Repository -> https://github.com/programmer-naman/Hypervision
The repository contains all the important backend code of the project,including the mini DBMS,SQL parser,query plan,scan,filter,projection,join,group and sort operation.It also include the lock manager,resource monitor,simulated VMI adapter and experiments manager.Important comits are related to adding these main features and testing the backend.The web dashboard and real Xen/LibVMI parts will be added in the next phase.

Testing and Validation Status

Test Type	Status (Pass/Fail)	Notes
SQL Parser and Query Test	Pass	SQL queries are parsed and run correctly.
DBMS Operations Test	pass	Scan,filter,join,group and sort is working.
Resource Monitoring Test	pass	CPU and memory usage is recorded properly.
Deadlock Detection Test	pass	Lock manager can detect deadlock.
Experiment and Rerun Test	pass	Query is run with different thread and result are compared.

Deliverables Progress

The main backend part of HyperVision is mostly completed.We have made the mini DBMS with SQL parser,query plan and different operation like scan,filter,projection,join,group and sort.The lock manager for detecting deadlock is also added.Resource monitor is used for checking CPU and memory usage,and a simulated VMI adapter is added for guest event.The experiment manager is also completed with experiment IDs,timeline,bottleneck classification,recomendations and thread based rerun.The backend is tested from the terminal using different queries and the results are saved in log files.Some important part are still remaining.The web dashboard is not completed yet,and real Xen with DomU and LibVMI integration is also pending.We also need to add more SQL support,run more experiments and improve the classifier.The final report and project demo is also remaining and will be completed in the next phase.